

Computación en Internet I

Semana 3

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Definiciones

Binarios

Subneting

NAT

IPv6

Definiciones

- Binary Numbers
- Network mask / Subnetwork Mask
- Network ID / Network Direction
- Host ID / Host Direcction
- Broadcast ID / Broadcast Direction
- Default Gateway

Binary table

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

IPv4 Classes

Class	First Range	Mask
A	0 - 127	/8
B	128 - 191	/16
C	192 - 223	/24
D	224 - 230	Multicast
E	240 - 255	Reserved

Private address range		
Class	start address	finish address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

Public address range		
Class	start address	finish address
A	0.0.0.0	126.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	254.255.255.255

IPv4 Decimal Format

Host

192.168.1.0/24

Network Mask

The diagram illustrates the IPv4 decimal format 192.168.1.0/24. The IP address is split into two parts: the network part (192.168.1) and the host part (0). The network part is labeled 'Network' and the host part is labeled 'Host'. The slash and the number 24 are labeled 'Mask'.

Binary Format

192.168.1.0/24

11000000.10101000.00000001.00000000

Network Host

Network Mask

192.168.1.0/24
Mask

(Decimal Format) 255.255.255.0

11111111.11111111.11111111.00000000

Subnetting

192.168.1.0/24

11000000.10101000.00000001.00000000

Network Sub Nets Host

Subnetting

192.168.1.0/28

11000000.10101000.00000001.00000000

Network Sub Nets Host

Subnetting

- $2^n - 2 = \text{Number of Host}$, donde n es la cantidad de bits que hay en el octeto de host
- $2^x = \text{Number of Subnets}$, donde x es la cantidad de bits que se toman del octeto de host para definir la subnet

NAT

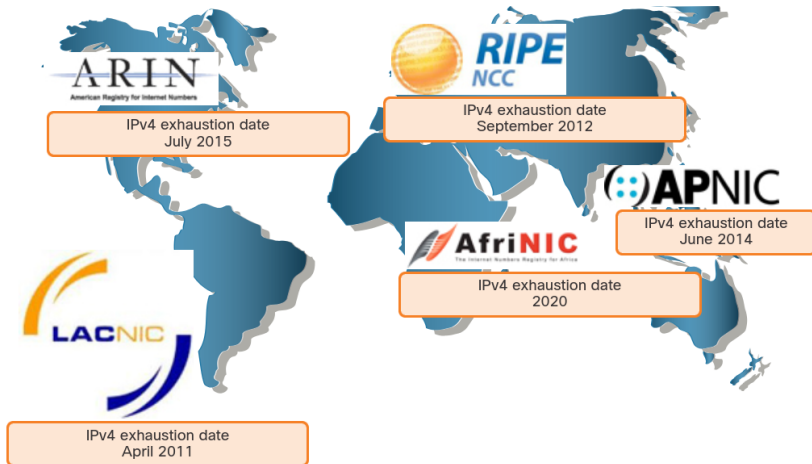
Network Address Translation

Hoja de ayuda

Subnetting IPv4

El problema

RIR - Regional Internet Registry



IPv4 vs IPv6		
Características	IPv4	IPv6
Tamaño de la dirección	32 bits	128 bits
Formato de la dirección	Decimal	Hexadecimal
Cantidad de Direcciones	2^{32}	2^{128}
Configuración	Manual o DHCP	Automatica (SLAAC)

IPv6

- Autoconfiguración
- Soportado desde 2003
- Falta de presión en el mercado
- Personal calificado
- Retorno de inversión

IPv6

- Redes de sensores
- Sistemas de transporte inteligente (ITS)
- Cloud Computing
- Monitoreo Remoto
- Obligatorio para la implementación de nuevas tecnologías [RFC6540, BCP0177]

Estado de IPv6

- MAPA del APNIC
- Estadísticas Google
- Colombia según LACNIC

IPv6

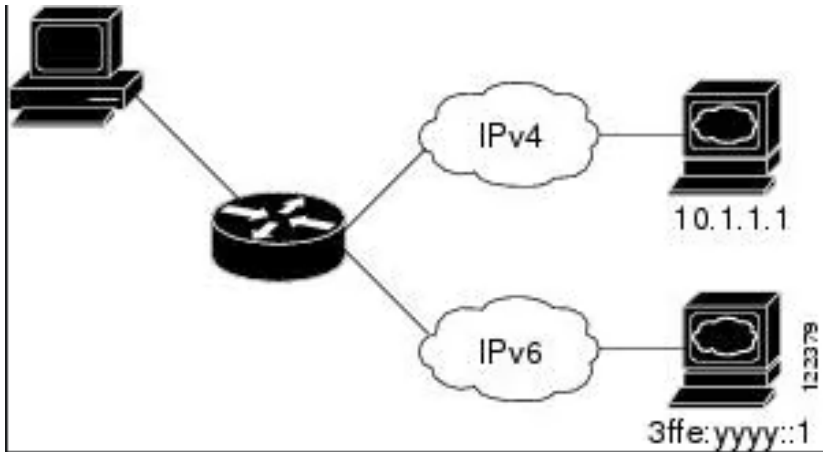
RFC 5952

- Dirección IP original =
2001:0DB8:00FF:0000:0008:0811:200C:417A
- Formato comprimido =
2001:DB8:FF:0:8:811:200C:417A
- Formato IP original =
2001:0DB8:0000:0000:0000:0000:0000:0043
- Formato comprimido = 2001:DB8::43

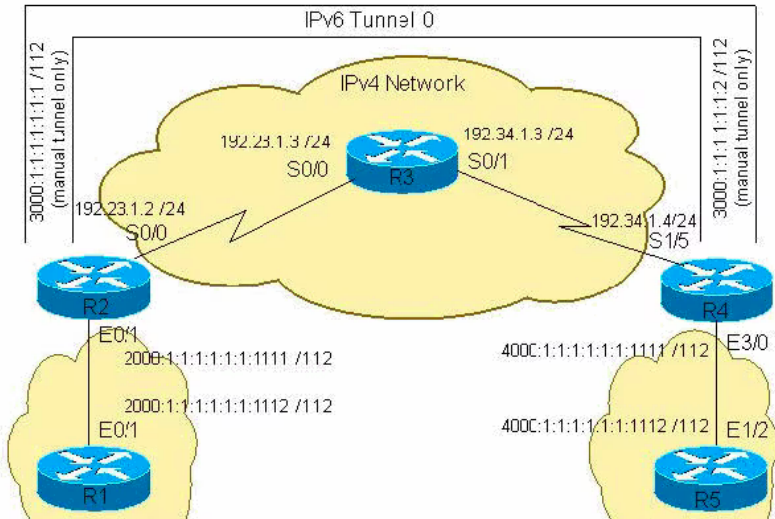
IPv4 and IPv6

- Dual Stack
- Tunneling
- Translation

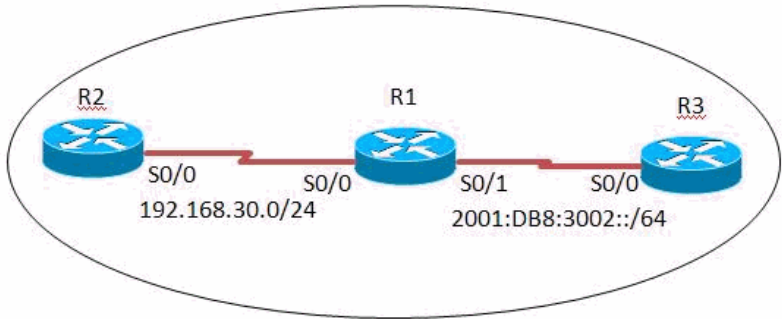
Dual Stack



Tunneling



Translation





Gracias